INCH-POUND

MIL-S-2063J
12 April 1989
SUPERSEDING
MIL-S-2063H
24 February 1984

MILITARY SPECIFICATION

SCARF, NECKWEAR, WOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of knitted wool scarf for neckwear.

1.2 Classification, The scarfs shall be one of the following classes as specified (see 6.2):

Class 1 - Olive Green 208
Class 2 - Gray 1155
Class 4 - Blue 3346

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8440

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(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, Durham, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

(Color cards may be available from the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3.3 Color. The color of the scarf shall be as specified (see 1.2).

3.3.1 Matching. The color of the finished fabric shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a corrulated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.3.2 Colorfastness. The dyed scarf shall show fastness to laundering, perspiration, light, and crocking equal to or better than the standard sample. When no standard sample is available, the dyed scarf shall show good fastness to laundering, perspiration, and light, and shall show an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5. Testing shall be as specified in 4.4.4.

3.4 Shrink resistant treatment. The wool shall be treated for resistance to felting shrinkage in stock, top, yarn, or fabric by a controlled oxidation process approved by the contracting activity. The shrink resistant process used shall not be identified by name or trademark, either on the scarf or on the package.

3.4.1 Alkali solubility. The alkali solubility of the shrink resistant treated wool for the finished scarf, in the form in which it has been treated (stock, top, yarn, or fabric) shall not be increased by more than 6.0 percent absolute) over the untreated material when tested as specified in 4.4.5.

3.5 Label. Each scarf shall have a combination size, identification and instruction label conforming to type VI, class 14 of DDD-L-20. Instructions shall be as follows:

DRY CLEAN OR HAND WASH
GENTLY IN LUKE WARM SUDS.
DO NOT TWIST OR WRING. DRY
ON FLAT SURFACE AWAY FROM
HEAT. DO NOT TUMBLE DRY.

3.6 Design. The scarf shall be tubular seamless type with reinforced ends.

3.7 Construction.

3.7.1 Stitches, seams and stitchings. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no run-offs, pleats, puckers or open seams occur. Thread breaks shall be backstitched not less than 1/2 inch.

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TABLE I.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. Cut tubing to the specified scarf length (see table II).						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Reinforce scarf ends. Place single layer of tape around inside of both ends of the scarf with ends of tape overlapped not less than 1/2 inch. The gage of stitching shall be 3/16 ± 1/16 inch.						
4.	Overstitching.						
	Overstitch ends of scarf.	504 or	SSa-1	12-16	50-3	Knitting yarns	Knitting yarns
		505	SSa-1	12-16	Knitting yarns	Knitting yarns	Knitting yarns
5.	Bartacking. Bartack the overlapping ends of tape to the scarf with a bartack.	Bartack		28 per bartack			

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

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3.8 Physical requirements.

3.8.1 Finished measurements and weights. The measurements and weights of the finished scarfs shall conform to table II. All measurements in table II shall be made with the scarf laid out flat under no tension.

TABLE II. Measurements and weights

		Tolerance
Courses per inch	10	Minimum
Wales per inch	8	Minimum
Width (in inches)	9	± 1/2
Length (in inches)	54	± 1-1/2
Weight (ounces per scarf)	6.0	Minimum

3.8.2 Bursting strength. The knitted fabric of the finished scarf shall have a bursting strength of not less than 60 pounds when tested as specified in 4.4.4.

3.8.3 Wool content. The knitted fabric of the finished scarf shall contain not less than 95 percent wool based on the dry weight of the specimen when tested as specified in 4.4.4.

3.8.4 Shrinkage requirements. The finished scarf shall not shrink or elongate more than 13.0 percent in either direction after one laundering cycle when tested as specified in 4.4.4.

3.9 Workmanship. The finished scarf shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.4.1.1 Wool grade. The wool shall be tested for grade in top form by comparison with the applicable U.S. Standard. In the event of a dispute, the wool grade shall be determined by the width (wedge) method. The sample unit shall be 1 yard of undyed (or untreated) top. The lot size shall be expressed in units of 1 pound. The lot shall be unacceptable if one or more units fail to meet any requirements specified. The sample size shall be as follows:

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of scarfs. The sample unit shall be one scarf. The inspection level shall be II and the acceptable quality level (AOL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE III. End item visual defects

Defect
Any hole, cut, tear, or dropped stitches.
Any mend.
Reinforcing tape omitted at either end or not applied to inside end of scarf.
Ends of tape not securely bartacked.
Broken or missing yarn over 1 inch in length.
Rolled, puckered, or ragged scarf ends.
Overedge stitching omitted, broken, skipped, incorrect seam type, or number of stitches less or more than specified.
Non-uniform in overall appearance, slubby yarn, or tiger stripes, clearly visible. 1/
Burr, shive, or sisal clearly visible. 1/
Knot not pulled through from face of fabric.
Face of fabric not lightly brushed.
Wrong color.
Shade bar, off shade, or stains clearly visible. 1/

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TABLE V. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method
Alkali solubility (stock, top, yarn fabric)	3.4.1	2800 2/
Texture:		
Courses per inch	3.8.1	5070
Wales per inch	3.8.1	5070
Weight of scarf	3.8.1	3/
Bursting strength	3.8.2	5120
Wool content	3.8.3	2101
Shrinkage (after 1 laundering cycle)	3.8.4	5556 4/

1/ One determination per sample unit and the results reported as "pass" or "fail".

2/ Sample unit for test shall consist of treated and untreated material as follows:
Stock - 1/4 pound, top - 1 yard, yarn - 1 cone, fabric - 1/4 yard.

3/ Prior to determination of any other test characteristic, the four scarfs encompassing the sample unit for testing shall be allowed to reach equilibrium under standard conditions as defined in FED-STD-191. After reaching equilibrium, the four scarfs shall be individually weighed to the nearest ounce and the average of the four determinations shall be reported to the nearest 0.1 ounce.

4/ The wool laundering procedure shall be used except the specimens shall be air dried (not tumble dried) and shall not be pressed prior to measuring.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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5.1.2 Commercial. Each scarf shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Seventy-five scarfs of one class only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, five in length, one in width and three in depth, within a shipping container. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Seventy-five scarfs of one class only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A units packs shall be packed flat, five in length, one in width, and three in depth within a shipping container. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have its contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Scarfs preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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6.5 Subject term (key word) listing.

Clothing
Cold weather

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 8440-0179)

Review activities:

Army - MD
Air Force - 82
DLA - CT

User activity:

Navy - MC
Air Force - 45

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-2063J	SCARF, NECKWEAR, WOOL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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